

1/4/23

Allen - Board Practice test - 2

PAGE No.

DATE

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Science - Biology

- A-1) (4) It passes neither through liver nor through pancreas
- A-2) (2) Living beings consume oxygen during respiration.
- A-3) (2) Bud develops due to repetitive cell division at a specific site.
- A-4) (4) Pisces only
- A-5) (4) Test tube B as pepsin can breakdown protein into simple molecules only in acidic conditions
- A-6) (2) Both A and R are true and R is not the correct explanation of A.
- A-7) (2) Both A and R are true and R is not the correct explanation of A.
- A-8) Rabbit and Deer belong to the same trophic level as both of them are primary consumers. Both of these animals consume only on grass and trees and are herbivores.

A-9) The de-oxygenated blood is found on the right side of heart i.e. in the right auricle and in the right ventricle. This blood is brought here by the superior and inferior vena cava from the whole body.

The oxygenated blood is found on the left side of heart i.e. in the left auricle and left ventricle. This blood is brought here by pulmonary vein from the lungs.

A-10) a) The contrasting character of Yellow seed is Green seed.

Out of these two the green seed is dominant and yellow seed is recessive.

b) The contrasting character of Round seed is wrinkled seed.

Out of these two the round seed is dominant and wrinkled seed is recessive.

A-11) a) In the tobacco plant male gamete has twenty four chromosomes and the female gamete also has twenty four chromosomes.

The number of chromosomes in the zygote are 48 i.e. 24 pairs.

b) i) Female child - XX

ii) Male Child - XY

A-12) i) The Pancreas secrete digestive enzyme (Trypsin & Lipase) and hormones (insulin) also.

ii) The hormone iodine is an important nutrient for our thyroid gland. The thyroid gland secretes the thyroxine hormone.

The thyroxine hormone controls the fat and protein metabolism in our body.

If the iodine intake is less in the body then it will hamper the production of thyroxine hormone. This can also cause the Goitre disease.

13) i) Mendel selected pea plant for his experiment because:-

1) It has a short lifespan, so he could study many generations.

2) It had seven, 7 pairs of contrasting characters.

ii) The trait which is able to express itself in F_1 generation is the dominant trait

The trait which is ~~ab~~ not able to express itself in the F_1 generation is called the recessive trait.

iii) The two visible traits that Mendel considered in his experiments are:-

1) Height of the plant i.e. Tall plant or Dwarf plant.

2) Colour of the flower i.e. White flower or purple flower.

A-14) a) i) The Ovary in a female is responsible for the gamete formation i.e. egg and also secretes the hormones like Oestrogen and progesterone.

ii) The oviduct or fallopian tube is the site where the fertilization of sperm and egg takes place.

b) i) The uterus when it receives the zygote is supplied by extensive blood network so that the zygote is provided by the nutrients.

ii) When the uterus does not receive any zygote takes out the excess blood along with some mucus through vagina.

in the form of a period.

Foetus

- c) The placenta connects the zygote to mother and helps to provide oxygen & water and nutrients to the foetus.